Chapter 1. Groups and Subgroups

Sets and relations, binary operations, isomorphisms, groups, subgroups, cyclic groups, generating sets

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These notes follow Fraleigh, *A First Course in Abstract Algebra*; the book's numbering is given in brackets.

1.1 Sets and relations (§0)

Motivation. Algebra studies sets equipped with operations, and almost every construction in these notes (cosets, factor groups, residue classes, fields of quotients) is a set of equivalence classes. Theorem 0.22 is the reason such constructions are legitimate: an equivalence relation is the same thing as a partition, so "identifying" equivalent elements always produces a well-defined new set.

A **relation** between sets A and B is a subset $\mathcal{R} \subset A \times B$; a **function** $\phi : A \to B$ is a relation in which each $a \in A$ appears as first member of exactly one pair (a, b), written $b = \phi(a)$. ϕ is **one-to-one** (injective) if $\phi(a_1) = \phi(a_2)$ implies $a_1 = a_2$, **onto** (surjective) if every b is a value, and a **bijection** if both. Sets have the same **cardinality** if there is a bijection between them. A **partition** of S is a collection of nonempty disjoint subsets (**cells**) whose union is S. A relation $\sim$ on S is an **equivalence relation** if it is reflexive ($x \sim x$), symmetric ($x \sim y \Rightarrow y \sim x$) and transitive ($x \sim y$, $y \sim z \Rightarrow x \sim z$).

Theorem 1.1 (0.22, equivalence relations and partitions)

Let $\sim$ be an equivalence relation on a nonempty set S. The sets $\bar{a} = \{x \in S : x \sim a\}$ form a partition of S. Conversely, each partition of S gives the equivalence relation "$x \sim y$ iff x, y lie in the same cell".

Proof. Each a lies in $\bar{a}$ (reflexivity), so the cells are nonempty and cover S. Suppose $c \in \bar{a} \cap \bar{b}$. If $x \in \bar{a}$, then $x \sim a$, $a \sim c$ (symmetry) and $c \sim b$, so $x \sim b$ (transitivity) and $\bar{a} \subset \bar{b}$; symmetrically $\bar{b} \subset \bar{a}$. So distinct cells are disjoint.

Conversely, every x is in a cell with itself; if x, y share a cell so do y, x; and if x, y share a cell and y, z share a cell, these cells contain y and hence coincide. ■

Example 1.1

Congruence modulo n: for $n \in \mathbb{Z}^+$, $h \equiv k \pmod{n}$ iff n divides $h - k$. It is an equivalence relation whose cells are the n **residue classes modulo** n.

1.2 Binary operations and isomorphisms (§§1–3)

Motivation. $\langle \mathbb{R}, + \rangle$ and $\langle \mathbb{R}^+, \cdot \rangle$ look different but behave identically once we match $x \leftrightarrow e^x$. Isomorphism makes "behaves identically" precise; from now on we care only about properties that survive isomorphisms, and the way to prove two structures different is to find such a structural property held by one but not the other.

A **binary operation** $*$ on S is a function $S \times S \to S$. It is **commutative** if $a * b = b * a$ and **associative** if $(a * b) * c = a * (b * c)$ for all a, b, c. A subset H is **closed** under $*$ if $a * b \in H$ for $a, b \in H$. Examples: $+$ and $\cdot$ on $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$; on the circle $U = \{z \in \mathbb{C} : |z| = 1\}$, multiplication; on $\mathbb{R}_c = [0, c)$, addition modulo c; on $U_n = \{z : z^n = 1\}$ and $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$, multiplication and addition modulo n.

Theorem 1.2 (2.13, associativity of composition)

For functions $f, g, h : S \to S$, $f \circ (g \circ h) = (f \circ g) \circ h$.

Proof. For $x \in S$, both sides send x to $f(g(h(x)))$. ■

A **binary algebraic structure** $\langle S, * \rangle$ is a set with a binary operation. An **isomorphism** $\phi : \langle S, * \rangle \to \langle S', *' \rangle$ is a bijection with $\phi(x * y) = \phi(x) *' \phi(y)$. To show two structures are isomorphic: define ϕ, show it is one-to-one and onto, and verify the homomorphism property. To show they are not, find a **structural property** (preserved by every isomorphism, such as cardinality, commutativity, or existence of solutions of $x * x = c$) held by one and not the other.

Example 1.2

$\phi(x) = e^x$ is an isomorphism $\langle \mathbb{R}, + \rangle \to \langle \mathbb{R}^+, \cdot \rangle$. $\langle \mathbb{R}_{2\pi}, +_{2\pi} \rangle \cong \langle U, \cdot \rangle$ via $x \mapsto e^{ix}$, and $\langle \mathbb{Z}_n, +_n \rangle \cong \langle U_n, \cdot \rangle$ via $k \mapsto e^{2\pi i k/n}$. $\langle \mathbb{Q}, + \rangle \ncong \langle \mathbb{R}, + \rangle$ (cardinality), $\langle \mathbb{Z}, + \rangle \ncong \langle \mathbb{Z}^+, + \rangle$ (in $\mathbb{Z}$, $x + x = c$ has a solution for $c = 0$), and $\langle \mathbb{R}, \cdot \rangle \ncong \langle \mathbb{C}, \cdot \rangle$ ($x \cdot x = c$ is solvable for every $c \in \mathbb{C}$).

An **identity element** e satisfies $e * s = s * e = s$ for all s; it is unique ($e = e * e' = e'$).

Theorem 1.3 (3.14)

If $\langle S, * \rangle$ has an identity e and $\phi : \langle S, * \rangle \to \langle S', *' \rangle$ is an isomorphism, then $\phi(e)$ is an identity for $*'$.

Proof. For $s' \in S'$ write $s' = \phi(s)$ (onto). Then $\phi(e) *' s' = \phi(e * s) = \phi(s) = s'$, and similarly $s' *' \phi(e) = s'$. ■

1.3 Groups (§4)

Motivation. Solving $a * x = b$ needs exactly three things: regrouping (associativity), a neutral element, and the ability to undo a (an inverse). A group is the abstraction of "an operation in which every equation $a * x = b$ can be solved uniquely"; numbers under addition, nonzero numbers under multiplication, and symmetries of an object all share this.

Definition (Group)

A **group** $\langle G, * \rangle$ is a set with a binary operation such that
$\mathcal{G}_1$: $*$ is associative;
$\mathcal{G}_2$: there is $e \in G$ with $e * x = x * e = x$ for all x;
$\mathcal{G}_3$: each a has an inverse a' with $a' * a = a * a' = e$.
It is **abelian** if $*$ is commutative.

Theorem 1.4 (4.15, cancellation laws)

In a group, $a * b = a * c$ implies $b = c$, and $b * a = c * a$ implies $b = c$.

Proof. Multiply $a * b = a * c$ on the left by a' and use associativity: $b = e * b = (a' * a) * b = a' * (a * b) = a' * (a * c) = c$. The right law is symmetric. ■

Theorem 1.5 (4.16)

In a group, the equations $a * x = b$ and $y * a = b$ have unique solutions x, y.

Proof. $x = a' * b$ is a solution, and any two solutions satisfy $a * x_1 = a * x_2$, hence are equal by Theorem 1.4. Similarly $y = b * a'$. ■

Theorem 1.6 (4.17)

In a group the identity element is unique, and each element has a unique inverse.

Proof. Identity: $e = e * e' = e'$. If a' and a'' are inverses of a, then $a * a' = e = a * a''$, so $a' = a''$ by cancellation. ■

We write a^{-1} for the inverse and, multiplicatively, a^n for powers ($a^0 = e$, $a^{-n} = (a^{-1})^n$); additively, $-a$ and na.

Corollary 1.7 (4.18)

$(a * b)' = b' * a'$.

Proof. $(a * b) * (b' * a') = a * (b * b') * a' = a * a' = e$, so $b' * a'$ is the inverse of $a * b$ by uniqueness. ■

Example 1.3

$\langle \mathbb{Z}, + \rangle$, $\langle \mathbb{Q}^*, \cdot \rangle$, $\langle \mathbb{Z}_n, +_n \rangle$, $\langle U_n, \cdot \rangle$, and $GL(n, \mathbb{R})$ (invertible $n \times n$ matrices under multiplication, nonabelian for $n \geq 2$) are groups; $\langle \mathbb{Z}^+, + \rangle$ and $\langle \mathbb{Z}, \cdot \rangle$ are not. Up to isomorphism there is exactly one group of each order $1, 2, 3$ (the group table is forced: each element appears exactly once in each row and column, by Theorem 1.5), and two groups of order 4: $\mathbb{Z}_4$ and the Klein 4-group V, in which every element is its own inverse.

1.4 Subgroups (§5)

Motivation. Just as one studies a vector space through its subspaces, a group is understood through the smaller groups sitting inside it. The two results here give a quick test for being a subgroup and the most important source of subgroups: the powers of a single element.

$H \subset G$ is a **subgroup**, $H \leq G$, if it is closed under the operation of G and is a group under it. The **order** $|G|$ of G is its number of elements.

Theorem 1.8 (5.14)

$H \subset G$ is a subgroup iff (1) H is closed under the operation, (2) $e \in H$, and (3) $a \in H \Rightarrow a^{-1} \in H$.

Proof. If $H \leq G$, (1) holds by definition. The identity e_H of H satisfies $e_H e_H = e_H = e_H e$ in G, so $e_H = e$ by cancellation, giving (2); the inverse of a in H is an inverse in G, hence equals a^{-1}, giving (3). Conversely (1)–(3) give $\mathcal{G}_2$ and $\mathcal{G}_3$ in H, and associativity is inherited from G. ■

For finite H, closure alone suffices (Exercise in the book: the powers of a repeat, so a^{-1} is a power of a). Equivalently, a nonempty H is a subgroup iff $ab^{-1} \in H$ for all $a, b \in H$.

Theorem 1.9 (5.17)

For $a \in G$, $H = \{a^n : n \in \mathbb{Z}\}$ is a subgroup of G, and it is the smallest subgroup containing a.

Proof. $a^r a^s = a^{r+s}$, $e = a^0$, $(a^r)^{-1} = a^{-r}$; apply Theorem 1.8. Any subgroup containing a contains all a^n by closure under products and inverses. ■

$H = \langle a \rangle$ is the **cyclic subgroup generated by** a; G is **cyclic** if $G = \langle a \rangle$ for some a, a **generator**. The **order** of a is $|\langle a \rangle|$.

1.5 Cyclic groups (§6)

Motivation. Cyclic groups are the simplest groups, generated by one element, and they can be classified completely: every cyclic group is $\mathbb{Z}$ or some $\mathbb{Z}_n$, and all their subgroups are cyclic again. They serve as the building blocks for the abelian groups of §11 and as the testing ground for later ideas such as order and index.

Theorem 1.10 (6.1)

Every cyclic group is abelian.

Proof. $a^r a^s = a^{r+s} = a^s a^r$. ∎

Theorem 1.11 (6.3, division algorithm for $\mathbb{Z}$)

If $m \in \mathbb{Z}^+$ and $n \in \mathbb{Z}$, there are unique $q, r \in \mathbb{Z}$ with $n = mq + r$ and $0 \le r < m$.

Proof. The set $\{n - mq : q \in \mathbb{Z}\} \cap \{0, 1, 2, \dots\}$ is nonempty (take $q = -|n|$), so has a least element $r = n - mq$. If $r \ge m$, then $r - m = n - m(q+1) \ge 0$ is smaller.

Uniqueness: if $mq + r = mq' + r'$ with $0 \le r, r' < m$, then m divides $r - r'$ and $|r - r'| < m$, so $r = r'$ and $q = q'$. ∎

Theorem 1.12 (6.6)

A subgroup of a cyclic group is cyclic.

Proof. Let $G = \langle a \rangle$ and $H \le G$. If $H = \{e\}$, $H = \langle e \rangle$. Otherwise H contains some a^n with $n > 0$ (if $a^{-n} \in H$, so is a^n); let m be the least such. For $b = a^n \in H$ write $n = mq + r$, $0 \le r < m$; then $a^r = a^n (a^m)^{-q} \in H$, so $r = 0$ by minimality, and $b = (a^m)^q$. Thus $H = \langle a^m \rangle$. ∎

Corollary 1.13 (6.7)

The subgroups of $\langle \mathbb{Z}, + \rangle$ are exactly the groups $n\mathbb{Z}$, $n \in \mathbb{Z}$.

Proof. $\mathbb{Z} = \langle 1 \rangle$; a subgroup generated by n is $n\mathbb{Z}$. ∎

For $r, s \in \mathbb{Z}^+$ the subgroup $\{nr + ms : n, m \in \mathbb{Z}\}$ is $d\mathbb{Z}$ for a unique $d > 0$, the **greatest common divisor** $\gcd(r, s)$: d divides r and s, every common divisor divides $d = nr + ms$, and d is a linear combination of r and s. r, s are **relatively prime** if $\gcd(r, s) = 1$. If r, s are relatively prime and r divides sm, then r divides m (multiply $1 = ar + bs$ by m).

Theorem 1.14 (6.10)

Let $G = \langle a \rangle$. If G is infinite, $G \cong \langle \mathbb{Z}, + \rangle$; if $|G| = n$, $G \cong \langle \mathbb{Z}_n, +_n \rangle$.

Proof. If all powers of a are distinct ($a^h = a^k$, $h > k$ would give $a^{h-k} = e$), then $a^m \mapsto m$ is a well-defined bijection $G \to \mathbb{Z}$, and $a^i a^j = a^{i+j}$ shows it is an isomorphism. Otherwise let n be the least positive integer with $a^n = e$. Then $e, a, \dots, a^{n-1}$ are distinct (else a smaller positive power is

e), and for any $m = nq + r$, $a^m = a^r$ (Theorem 1.11); so $G = \{a^0, \ldots, a^{n-1}\}$, $|G| = n$, and $a^i \mapsto i$ is an isomorphism with $\mathbb{Z}_n$, since $a^i a^j = a^{i+_n j}$. ■

In particular, if a has finite order n, then $a^m = e$ iff n divides m.

Theorem 1.15 (6.14)

Let $G = \langle a \rangle$ have order n and $b = a^s$. Then b generates a cyclic subgroup of order n/d, $d = \gcd(n, s)$. Also $\langle a^s \rangle = \langle a^t \rangle$ iff $\gcd(s, n) = \gcd(t, n)$.

Proof. $|\langle b \rangle|$ is the least $m > 0$ with $a^{sm} = e$, i.e. with $n \mid sm$. Write $n = dn'$, $s = ds'$ with $\gcd(n', s') = 1$; then $n \mid sm$ iff $n' \mid s'm$ iff $n' \mid m$. So the order is $n' = n/d$. For the last statement: in the cyclic group $\mathbb{Z}_n$, $\langle s \rangle$ is the unique subgroup of order n/d (it is $\langle d \rangle$: $d \in \langle s \rangle$ as $d = xs + yn$, and $\langle d \rangle$ has order n/d); so $\langle a^s \rangle = \langle a^{\gcd(s,n)} \rangle$, and distinct divisors d of n give subgroups of distinct orders. ■

Corollary 1.16 (6.16)

If a generates a cyclic group of order n, the other generators are the a^r with r relatively prime to n.

Proof. a^r generates iff its order $n/\gcd(r, n)$ is n. ■

Consequently a cyclic group of order n has exactly one subgroup of each order d dividing n, and $\phi(n)$ generators.

1.6 Generating sets and Cayley digraphs (§7)

Motivation. Few groups are cyclic, but most groups we meet are generated by a handful of elements. Describing a group by generators is economical, and the Cayley digraph turns the multiplication table into a picture in which a group element is a path.

Theorem 1.17 (7.4)

The intersection of any family of subgroups H_i $(i \in I)$ of G is a subgroup of G.

Proof. If $a, b \in \bigcap H_i$, then ab and a^{-1} lie in every H_i, as does e; apply Theorem 1.8. ■

For $\{a_i : i \in I\} \subset G$, the intersection of all subgroups containing every a_i is the smallest such subgroup, the **subgroup generated by** $\{a_i\}$; if it is G, the a_i **generate** G.

Theorem 1.18 (7.6)

The subgroup H generated by $\{a_i : i \in I\}$ consists exactly of the finite products of integral powers of the a_i (a given a_i may occur several times).

Proof. Let K be the set of such finite products. K is closed under multiplication (juxtapose), contains $e = a_i^0$, and contains inverses: $(a_{i_1}^{n_1} \cdots a_{i_k}^{n_k})^{-1} = a_{i_k}^{-n_k} \cdots a_{i_1}^{-n_1}$. So $K \leq G$ contains each a_i, whence $H \subset K$. Conversely any subgroup containing the a_i contains all such products, so $K \subset H$. ■

A **Cayley digraph** of G for a generating set S has the elements of G as vertices and an arc of type s from g to gs for each $s \in S$. It is connected; at each vertex exactly one arc of each type starts

and one ends; and two different paths from a vertex g ending at the same vertex describe equal products in G.

Example 1.4

$V = \{e, a, b, c\}$ is generated by $\{a, b\}$ but is not cyclic. $\mathbb{Z}_6$ is generated by $\{2, 3\}$, as $1 = 3 - 2$. The dihedral-like group with digraph two concentric triangles joined by spokes (generators a of order 3 and b of order 2 with $ba = a^2b$) is S_3.